

15-Minute Demo Plan

Requirements Gap Analysis Agent in PyCharm

Message for Kiran

In our 15-minute call, I will show a working product-grade AI pipeline that analyzes large business and engineering transcripts, finds requirement coverage gaps, and generates a verified report.

What I Will Show

- A PyCharm-ready Python project for requirements gap analysis.
- Qwen3:8B through Ollama for local LLM reasoning.
- BAAI/bge-m3 for open-source semantic embeddings.
- Large transcript mode with chunking, retrieval, focused analysis, and report generation.

Why This Is Useful

Business teams describe requirements in meetings, while engineering teams describe implementation decisions separately. The product compares both sides and identifies what is covered, partially covered, deferred, or missing.

End-to-End Pipeline

Stage	What happens
1. Load	Read business and engineering transcripts.
2. Chunk	Split large transcripts into overlapping chunks to avoid context overflow.
3. Extract	Use Qwen to extract requirements and engineering solutions as structured JSON.
4. Merge	Deduplicate repeated items from overlapping chunks.
5. Retrieve	Use BGE-M3 embeddings to retrieve top-k candidate solutions for each requirement.
6. Analyze	Use Qwen for focused gap analysis over only relevant evidence.
7. Verify	Use Qwen critic verification to reduce false positives.
8. Report	Write final Markdown or JSON gap reports.

What Makes It Product-Grade

- Config-driven model selection and local open-source execution.
- Modular layers: schemas, parsers, chunking, agents, retrieval, merge, reporting.
- Tests for chunking, deduplication, retrieval, LLM parsing, and pipeline behavior.
- Scalable design: current local vector index can later be replaced by Qdrant or Supabase pgvector.

Demo Command

```
python main.py --mode large --config configs/qwen_local.yaml --business transcripts/eval_business
--engineering transcripts/eval_engineering --output piyush/eval_large_qwen.md --format markdown
```

Main takeaway

This is not just a prompt. It is an end-to-end AI system with transcript processing, open-source LLM extraction, embedding retrieval, focused reasoning, critic verification, and report generation.